

Comparative Image Restoration using K-SVD and DWT

CS342 - Image Processing Applications
Bonus Project

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Problem Overview

$$g(x, y) = f(x, y) + \eta(x, y) \quad (1)$$

- ▶ $f(x, y)$: Original image
- ▶ $\eta(x, y)$: Gaussian noise
- ▶ Goal: Recover clean image from noisy input

Problem Overview

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Approaches:

- ▶ DWT (Wavelet-based)
- ▶ K-SVD (Sparse representation)

Methods Overview

DWT (Wavelet Transform)

- ▶ Decomposes image into LL, LH, HL, HH
- ▶ Removes noise via thresholding

K-SVD (Sparse Representation)

- ▶ Works on image patches
- ▶ Learns dictionary for reconstruction

Role of Alpha (α)

- Controls denoising strength inside each method

In DWT:

$$T = \alpha \cdot \max(|C|) \quad (2)$$

In K-SVD:

$$\min ||Y - DX||^2 + \alpha ||X||_0 \quad (3)$$

Pipeline

1. Input Image
2. Add Gaussian Noise
3. Apply:
 - ▶ DWT Denoising
 - ▶ K-SVD Denoising
4. Compare Results

Comparison

Feature	DWT	K-SVD
Domain	Frequency	Spatial
Speed	Fast	Slower
Detail	Moderate	High

- ▶ DWT: good noise removal
- ▶ K-SVD: better detail preservation

Demo: Input & Noise Impact

Controls

IMAGE INPUT

Upload an image

Drag and drop file here
(Max 10240 per file; PNG, JPEG, ...)

Browse files

Screenshot 2024-04-17-20:20

NOISE PARAMETERS

Noise Level (σ)

STORAGE PARAMETERS

Alpha (α)

Input Images

Original Image (Grayscale)

Noisy Image (σ = 30)

Size: 300×200 px

Noise model: $g(x,y) = f(x,y) + n$, $n \sim N(0, 30^2)$

Noise Impact Metrics

PSNR	MSE	SSIM
21.22 dB	490.46	0.3855

Restoration Outputs

Alpha (α = 0.50) controls denoising strength inside each method independently

DWT = Threshold T = 0.50 × max(coefficients) = 207.0

R-SVD = max 2 non-zero atoms per patch (of 8×64)

- ▶ Gaussian Noise added ($\sigma = 30$)
- ▶ Initial PSNR: 21.22 dB
- ▶ Demonstrates degradation level

Demo: Restoration Outputs

Controls

IMAGE INPUT

Upload an image

Drag and drop file here
Limit 200MB per file • PNG, JPG, JPEG...

Browse files

 Screenshot 2026-04-...
17.3KB

NOISE PARAMETERS

Noise Level (α)

30

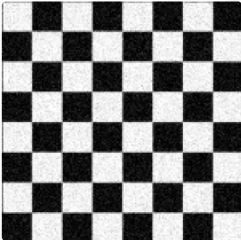
RESTORATION PARAMETERS

Alpha (α)

0.90

Deploy

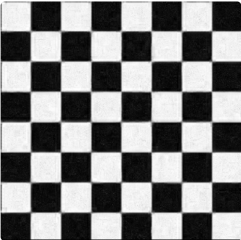
** DWT Denoised ($\alpha = 0.90$) **



Wavelet: db1 | T = $\alpha \cdot \max(c) = 0.90 \cdot 230 = 207.00$ | Higher $\alpha \rightarrow$ stronger thresholding

Download DWT Result

** K-SVD Denoised ($\alpha = 0.90$) **



Atoms: 64 | Active: 2 = $\max(1, \text{round}((1 - 0.90) \times 8^2 / 4))$ | Higher $\alpha \rightarrow$ fewer atoms $\rightarrow$ sparser

Download K-SVD Result

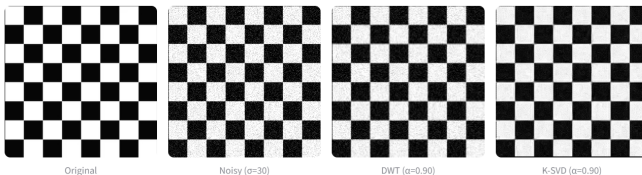
- ▶ DWT: Smooths noise using thresholding
- ▶ K-SVD: Preserves structure using sparse patches
- ▶ $\alpha = 0.90$ controls denoising strength internally

Demo: Comparative Analysis

PSNR 20.32 dB	MSE 604.34	SSIM 0.5239	PSNR 21.71 dB	MSE 438.13	SSIM 0.6749
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Comparative Analysis

Visual Comparison



- ▶ K-SVD achieves better PSNR and SSIM
- ▶ DWT removes noise but blurs edges

Demo: Difference Maps

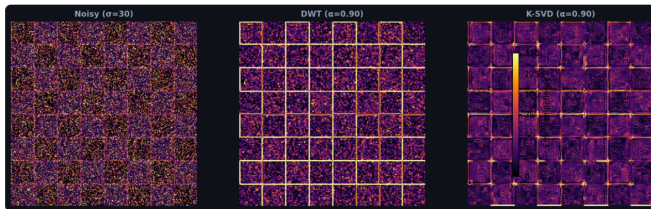
Numerical Comparison

Metric	Noisy	DWT	K-SVD	Winner
PSNR (dB) $\uparrow$	21.22	20.32	21.71	K-SVD
MSE $\downarrow$	490.46	604.34	438.13	K-SVD
SSIM $\uparrow$	0.3855	0.5239	0.6749	K-SVD

$\uparrow$ = higher is better | $\downarrow$ = lower is better | green = winner for that metric

Difference Maps |Original - Restored|

Bright regions indicate large restoration error. Differences are amplified 3 $\times$ for visibility.

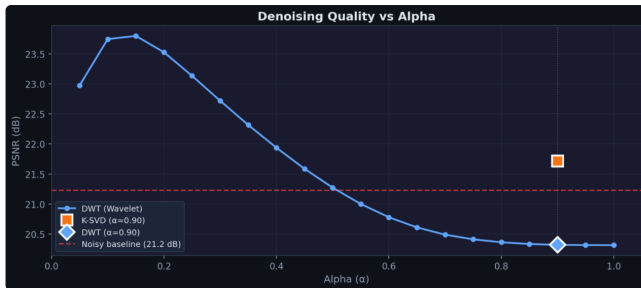


- ▶ Bright regions indicate higher error
- ▶ K-SVD shows lower reconstruction error

Demo: PSNR vs Alpha

PSNR vs Alpha (α)

DWT: full sweep (fast) | K-SVD: current α point (full sweep is computationally expensive)



Reading the graph: The DWT curve shows how PSNR changes across all α values. The K-SVD point shows quality at the current α . Both methods should peak at some optimal α — too low preserves noise, too high removes detail.

- Shows effect of α on denoising quality
- Optimal α lies in mid-range

Demo: DWT Subbands

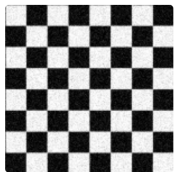
▼ DWT Subband Visualization (Educational)

The DWT decomposes the image into **four subbands** (Gonzalez, Ch. 7):

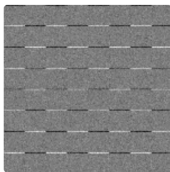
- LL — Approximation (low-frequency structure)
- LH — Horizontal details (horizontal edges)
- HL — Vertical details (vertical edges)
- HH — Diagonal details (noise + diagonal edges)

Soft thresholding is applied to LH, HL, HH only — the LL subband is preserved to maintain image structure.

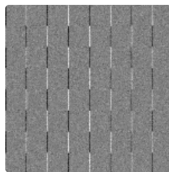
Before Thresholding (Noisy Subbands)



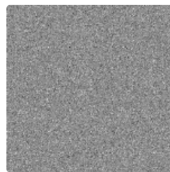
LL (Approx)



LH (Horiz)



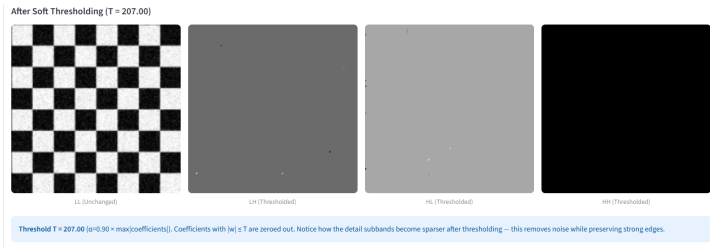
HL (Vert)



HH (Diag)

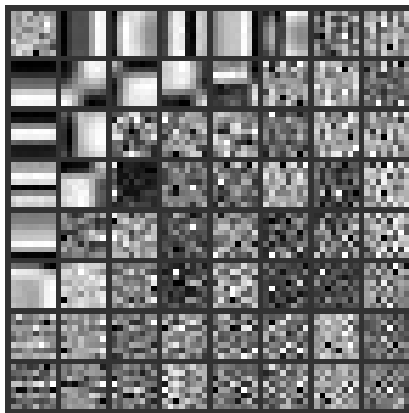
- ▶ LL: Structure
- ▶ LH, HL, HH: Detail + Noise

Demo: After Thresholding



- ▶ High-frequency noise removed
- ▶ Some details also lost

Demo: K-SVD Dictionary



- ▶ Learned atoms represent image structures
- ▶ Sparse combination reconstructs patches

Demo: Current Parameters

Current Parameters

Noise α

30

Alpha α

0.90

Patch Size

8×8

Iterations

10

How Alpha (α) Works in Each Method

DWT (Wavelet Thresholding)

Formula: $T = \alpha * \max(|\text{detail1_coefficients}|)$

- α directly scales the threshold T
- **Low α (0.1)** $\rightarrow$ small $T \rightarrow$ few coefficients zeroed $\rightarrow$ **mild denoising**
- **High α (1.0)** $\rightarrow$ large $T \rightarrow$ many coefficients zeroed $\rightarrow$ **strong denoising**

Current: $T = 0.90 * 230 = 207.0$

K-SVD (Sparse Representation)

Formula: $\text{max_atoms} = \max(1, \text{round}((1 - \alpha) * \text{patch_size}^2 / 4))$

- α inversely controls how many dictionary atoms each patch can use
- **Low α (0.1)** $\rightarrow$ many atoms $\rightarrow$ detailed reconstruction $\rightarrow$ **mild denoising**
- **High α (1.0)** $\rightarrow$ few atoms $\rightarrow$ coarse reconstruction $\rightarrow$ **strong denoising**

Current: $\text{max_atoms} = \max(1, \text{round}[(1 - 0.90) * 8^2 / 4]) = 2$

Important: Alpha acts **INSIDE** each method independently. It is **NOT** used to compare methods directly — it controls denoising strength within each algorithm.

Conclusion

- ▶ Both methods effectively remove noise
- ▶ K-SVD preserves textures better
- ▶ DWT is faster and simpler
- ▶ α plays a critical role

Thank You!